

NOVA INSIGHT

Predictive Schedule Intelligence Layer

Development Specification Report

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Deadline for prototype: Wednesday 15:00 Bali Time

Project: Nova Insight – Construction Schedule Intelligence

1. Executive Summary

Over the last weeks I have had several discussions with senior project leaders in the construction industry. A recurring request has emerged from multiple professionals, including senior planners and project directors working on large-scale projects such as the National Bank project in Denmark.

They all expressed a clear need for a system capable of analyzing complex construction schedules and predicting potential risks and inconsistencies before they cause delays.

This predictive layer is extremely valuable.

Multiple project managers have independently requested this functionality and confirmed that a system capable of detecting schedule anomalies and predicting issues based on a single schedule would be **gold for the industry**.

Construction schedules today often contain hundreds or thousands of activities. They are extremely difficult to interpret manually. Project managers spend significant time trying to detect issues that could be automatically identified by a system.

Nova Insight should therefore evolve beyond simple schedule comparison and introduce a **Predictive Schedule Intelligence Layer**.

This report defines exactly how that layer should be built.

2. The Core Industry Problem

Large construction projects rely heavily on schedule management tools such as:

- Microsoft Project
- Primavera P6
- Asta Powerproject

These systems produce complex schedules but do **not provide real intelligence**.

They display data but they do not analyze it.

Typical schedules contain:

- 800–2000 activities
- thousands of dependencies
- overlapping disciplines
- unrealistic progress reporting
- decision bottlenecks

The result is that project managers struggle to identify:

- which tasks are actually late
- which tasks are unrealistic
- which dependencies will cause delays
- which activities block others
- which decisions are missing

In the example schedule analyzed (National Bank project planning schedule), the following problems appear:

- Hundreds of micro-activities
- Many 0-day coordination tasks
- Large dependency clusters
- Multiple disciplines overlapping
- Activities with unrealistic progress values
- Long tasks with unclear progress
- Decision tasks hidden inside technical tasks

Without advanced analysis, it is nearly impossible to detect risks early.

This is where Nova Insight becomes extremely valuable.

3. Objective of the Predictive Layer

The objective of this system is to automatically analyze construction schedules and detect:

1. Activities that should already have started

2. Activities whose reported progress is unrealistic
3. Activities that block other activities
4. Risk clusters inside dependency chains
5. Decision bottlenecks
6. Artificial scheduling clusters
7. Potential delay risk across the schedule

Instead of presenting a complex Gantt chart, Nova should extract **clear insights**.

Example output:

"7 activities should have started but have 0% progress."

"Electrical installation chain is likely to delay lighting installation."

"Progress reporting on 12 activities appears unrealistic."

"4 tasks are blocking downstream work."

4. Data Structure Requirements

Nova must be able to read and parse the following schedule fields.

Minimum dataset required:

Activity ID

Activity Name

Duration

Start Date

Finish Date

Percent Complete

Predecessor Activities

Successor Activities

Example structure:

Activity ID

Task Name

Duration (days)

Start Date

Finish Date

% Complete

Predecessors

Successors

The system must be capable of ingesting data from:

PDF schedule exports

Excel exports

Primavera or MS Project exports

5. Detection Algorithms

The predictive system should include several analytical modules.

5.1 Detection Module A

Activities That Should Have Started

This was specifically requested by project directors.

Logic:

If an activity's start date is earlier than today and progress is still zero, it should be flagged.

Pseudo logic:

If

Start Date < Today

AND

Percent Complete = 0

Then flag activity.

Output example:

Activity: Electrical Installation Level 3

Planned Start: March 1

Progress: 0%

Status: Should have started.

5.2 Detection Module B

Unrealistic Progress Reporting

Project managers often report incorrect progress percentages.

Example:

Task duration: 100 days

Elapsed time: 28 days

Reported progress: 60%

Actual expected progress should be approximately 28%.

Calculation:

Expected Progress =

$(\text{Today} - \text{Start Date}) / \text{Duration}$

If difference between expected and reported progress exceeds threshold:

Example threshold:

20–30%

Then flag activity.

Example output:

Task: Cable Installation

Expected progress: 28%

Reported progress: 60%

Status: Progress anomaly.

5.3 Detection Module C

Dependency Chain Analysis

Construction schedules rely heavily on dependencies.

Example:

Cable trays → electrical installation → lighting → controls.

Nova must detect long chains of dependencies and analyze risk propagation.

If a task early in the chain is delayed, downstream tasks will also be affected.

Algorithm steps:

1. Build dependency graph

2. Identify chains of dependencies
3. Detect chains longer than threshold
4. Evaluate delay impact

Output example:

Critical dependency chain detected:

Electrical installation → lighting installation → controls commissioning.

Risk level: High.

5.4 Detection Module D

Decision Activities

Many activities have duration of zero days.

These are typically coordination or decision tasks.

Examples:

Approval of equipment

Component selection

Client decisions

These activities are often hidden schedule risks.

Logic:

If duration = 0 days

AND activity name contains keywords like:

approval

selection

decision

placement

coordination

Then classify as decision activity.

Output example:

Client decision required: Placement of fire safety components.

Risk level: medium.

5.5 Detection Module E

Artificial Scheduling Clusters

In many schedules multiple tasks start on exactly the same date.

Example found in the schedule:

9 tasks starting on the same day.

This often indicates placeholder planning rather than realistic scheduling.

Logic:

If more than X tasks share the same start date within the same discipline:

Flag cluster.

Example output:

Scheduling cluster detected.

8 tasks scheduled to start on Feb 9.

Possible planning placeholder.

5.6 Detection Module F

Long Duration Activities

Tasks exceeding a certain duration often carry higher risk.

Example thresholds:

90 days

120 days

If duration exceeds threshold:

Flag for risk.

Example output:

Task: Project Engineering Completion

Duration: 125 days

Risk level: Elevated.

5.7 Detection Module G

Discipline Clustering

The schedule contains repeated task types such as:

Electrical installation

Lighting installation

IT and low voltage

Cable trays

Nova should group tasks by discipline automatically.

This allows progress tracking per discipline.

Example output:

Electrical progress: 58%

Lighting progress: 42%

IT systems progress: 61%

6. Schedule Complexity Scoring

Nova should compute a complexity score.

Inputs:

Number of activities

Number of dependencies

Average chain length

Example output:

Activities: 1243

Dependencies: 2800

Average dependency chain: 7

Schedule complexity score: High.

This helps identify schedules that require deeper monitoring.

7. Predictive Delay Engine

This is the most important component.

The system should estimate delay risk using:

Activity progress
Dependency chains
Remaining durations
Blocked activities

Example output:

Potential delay risk: 34%

Estimated delay window: 18–24 days

Primary risk source:

Electrical installation chain.

8. User Output Layer

Nova should not overwhelm users with data.

The interface should present insights such as:

Schedule Health Overview

- 7 activities should have started
- 12 progress anomalies
- 4 blocked activities
- 2 critical dependency chains

Risk level: Medium.

9. Development Priority

This predictive layer must be implemented immediately.

Reason:

Multiple senior project managers have now requested this capability.

It solves a real industry problem and differentiates Nova from traditional scheduling software.

This feature is extremely valuable and has been described by industry professionals as **gold for project management**.

10. Delivery Deadline

Prototype implementation deadline:

Wednesday

15:00 Bali Time

Minimum deliverable:

Working predictive engine capable of analyzing:

Start dates

Durations

Percent progress

Dependencies

And producing anomaly reports.

11. Strategic Importance

If implemented correctly, this predictive layer will position Nova Insight as a true **schedule intelligence platform**, not just a schedule viewer.

This is the direction the product must evolve.

It will provide enormous value to:

Project directors

Planning engineers

Claims specialists

Client representatives